BIDIRECTIONAL CHARGER FOR A SINGLE-PHASE ELECTRIC VEHICLE USING BUCK-BOOST CONVERTER

Project Report

Submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor of Technology

in

Electrical Engineering

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May 2026

CERTIFICATE

This is to certify that the report entitled **"DESIGN AND DEVELOPMENT OF BIDIRECTIONAL EV CHARGER"** is a bonafide record of the Project done by **Ravi** (Roll No**.: B230694EE) , Rishikesh (**Roll No**.: B230522EE) , V Johnson (**Roll No.: **B231307EE)** , **Zukku Shiva Naga Mohan Reddy (**Roll No.: **B230716EE)** under our supervision, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Electrical and Electronics Engineering** from **National Institute of Technology Calicut**, and this work has not been submitted elsewhere for the award of a degree.

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our project guide, **Dr. Korra Balu**, Associate Professor, Department of Electrical and Electronics Engineering, NIT Calicut, for his invaluable guidance, constant encouragement, and expert advice throughout the course of this project. His insightful suggestions and constructive feedback have been instrumental in shaping the direction and quality of this work.

We are grateful to the Department of Electrical and Electronics Engineering and the National Institute of Technology Calicut for providing us with the necessary computing resources, laboratory facilities, and academic environment to carry out this research.

Finally, we are grateful to our families and peers for their unwavering support and encouragement throughout the course of this project.

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ABSTRACT

The rapid adoption of electric vehicles (EVs) has created a pressing need for efficient and intelligent charging infrastructure. This project investigates the design, analysis, and simulation of a bidirectional on-board single-phase EV charger capable of operating in both Grid-to-Vehicle (G2V) and Vehicle-to-Grid (V2G) modes using a Buck-Boost converter topology.

The proposed system employs a two-stage power conversion architecture: an AC-DC bidirectional converter interfacing with the single-phase utility grid (230 V, 50 Hz), and a DC-DC Buck-Boost converter managing energy exchange with the EV battery (nominal 160 V). In G2V mode, the system operates in buck mode, stepping down the 400 V DC link voltage to charge the battery at 160 V with a 23% duty cycle. In V2G mode, the system operates in boost mode, stepping up the battery voltage from 160 V to 400 V for grid injection, employing a 80% duty cycle. The rated charging power of the system is 2 kW.

A Phase-Locked Loop (PLL)-based control strategy synchronized with the grid is implemented to enable smooth transitions between operating modes. The battery State of Charge (SoC) is monitored to switch between Constant Current (CC) and Constant Voltage (CV) charging modes, with CC mode active below 80% SoC and CV mode above. An LCL filter is employed on the grid side to suppress harmonic distortion and ensure a pure sinusoidal AC output during V2G operation.

The complete system is modelled and simulated using MATLAB/PSIM. Simulation results validate the converter's ability to regulate DC bus voltage at 400 V, achieve stable battery charging at 160 V with 16 A output current, and successfully transfer power back to the grid during V2G operation. The results confirm that bidirectional EV chargers can contribute to peak load shaving, voltage regulation, and improved power system stability, making them a viable solution for smart grid integration.

Keywords: Bidirectional Charger, Electric Vehicle (EV), Grid-to-Vehicle (G2V), Vehicle-to-Grid (V2G), Buck-Boost Converter, State of Charge (SoC), LCL Filter, MATLAB/PSIM Simulation

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LIST OF ABBREVIATIONS

Acronym	
AC	Alternating Current
CC	Constant Current
CV	Constant Voltage
DC	Direct Current
EV	Electric Vehicle
G2V	Grid-to-Vehicle
IGBT	Insulated Gate Bipolar Transistor
LCL	Inductor-Capacitor-Inductor Filter
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor
PLL	Phase-Locked Loop
PSIM	Power Electronics Simulation
PWM	Pulse Width Modulation
SoC	State of Charge
THD	Total Harmonic Distortion
V2G	Vehicle-to-Grid

LIST OF SYMBOLS

Symbols

V_{grid}	Grid Voltage	Vrms
f_{grid}	Grid Frequency	Hz
P	Rated Charging Power	kW
V_{dc}	DC Link Voltage	Volt
f_{sw}	Switching Frequency	kHz
$V_{battery}$	Battery Voltage	Volt
I_o	Output Current	Amp
Cout	Output Capacitor	µF
L_1	LCL Filter Inductor 1	mH
L_2	LCL Filter Inductor 2	mH
C_f	LCL Filter Capacitor	µF
L_{buck}	Buck Converter Inductor	mH
C_{buck}	Buck Converter Capacitor	µF
D	Duty Cycle	—
D_{buck}	Buck Mode Duty Cycle	—
D_{boost}	Boost Mode Duty Cycle	—
ΔI_L	Inductor Current Ripple	Amp
ΔV_o	Output Voltage Ripple	Volt
θ_{grid}	Grid Phase Angle	rad
I_{ref}	Reference Charging Current	Amp
i_{bref}	Battery Current Reference	Amp
Q1, Q2	Converter Switching Devices	—
SoC	State-of-Charge	%
THD	Total Harmonic Distortion	%

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CHAPTER 1

INTRODUCTION

1.1 Background

The global transportation sector is undergoing a paradigm shift as concerns over carbon emissions from fossil fuel combustion mount. Conventional internal combustion engine (ICE) vehicles are a significant contributor to greenhouse gas emissions and urban air pollution. In response, electric vehicles (EVs) have emerged as a clean, sustainable alternative, offering zero tailpipe emissions and the potential for integration with renewable energy sources.

The widespread deployment of EVs, however, raises important challenges for the power grid. Large-scale EV charging, if uncoordinated, can impose significant stress on distribution networks, causing peak demand spikes and voltage instability. This challenge has spurred interest in smart, bidirectional EV charging technologies that not only draw power from the grid (G2V) but can also return stored energy to the grid (V2G), effectively transforming EV batteries into distributed energy storage assets.

A bidirectional on-board charger (OBC) is the enabling hardware for V2G capability. Unlike conventional unidirectional chargers that only convert AC grid power to DC for battery charging, a bidirectional OBC can also invert DC battery power to AC for grid injection. This requires a sophisticated two-stage power conversion system consisting of a bidirectional AC-DC converter (rectifier/inverter) and a bidirectional DC-DC converter (buck-boost) as illustrated in **Fig. 1.1**.

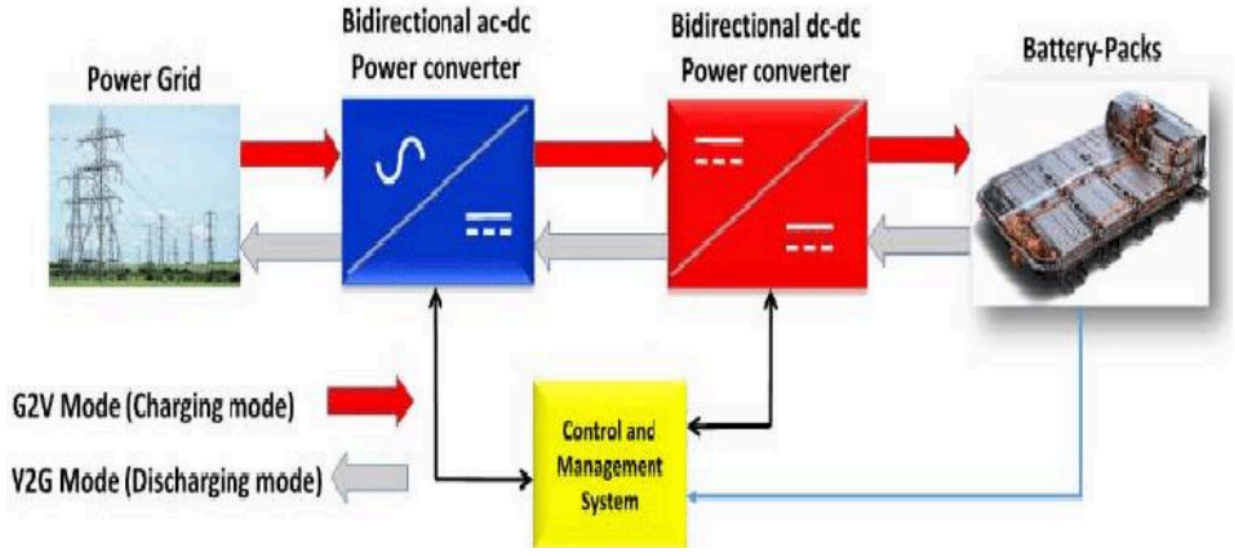


Fig. 1.1. Block diagram of a single-phase bidirectional charger system.

1.2 Motivation of the Current Research Project

The motivation for this project arises from the dual challenge of increasing EV adoption and managing the resulting grid impact. Bidirectional chargers offer a compelling solution: when millions of EVs are connected to the grid, their collective battery capacity represents a vast, distributed energy storage resource. Intelligently managed V2G systems can provide services such as:

- Peak load shaving by discharging EVs during periods of high grid demand
- Load levelling by charging EVs during off-peak hours and discharging during peak hours
- Frequency regulation and voltage support at the distribution level
- Emergency backup power for homes and critical infrastructure
- Integration of variable renewable energy sources like solar and wind

This project contributes to this field by designing and validating a single-phase bidirectional OBC using a Buck-Boost converter, demonstrated through MATLAB/PSIM simulation, with a particular focus on the control strategy for seamless G2V and V2G operation.

1.3 Objectives

The primary objectives of this project are:

(a) To design a single-phase bidirectional on-board EV charger capable of G2V and V2G operation.
(b) To analyse and implement a Buck-Boost converter topology for bidirectional DC-DC power conversion between the battery (160 V) and DC link (400 V).
(c) To develop a control algorithm incorporating Constant Current (CC) and Constant Voltage (CV) modes for safe battery charging, governed by the State of Charge (SoC).
(d) To design and evaluate an LCL filter to ensure power quality at the grid interface during V2G operation.
(e) To simulate the complete system in MATLAB/PSIM and validate G2V and V2G performance through waveform analysis.
(f) To assess the feasibility of the proposed system for smart grid integration and distributed energy storage applications.

1.4 Outline of the Report

The remainder of this report is organized as follows. **Chapter 2** describes the bidirectional charger topology including the two-stage power conversion architecture and the principles of buck and boost mode operation. **Chapter 3** presents the detailed system parameters and the analytical design equations for the converter components. **Chapter 4** describes the control system design including the CC/CV battery charging strategy,

the V2G power control loop, and the Phase-Locked Loop used for grid synchronization. **Chapter 5** presents and discusses the MATLAB/PSIM simulation results for both G2V and V2G modes of operation. **Chapter 6** summarizes the conclusions drawn from this work and identifies directions for future research.

CHAPTER 2

BIDIRECTIONAL CHARGER TOPOLOGY

2.1 System Architecture

The proposed single-phase bidirectional EV charging system consists of two cascaded power conversion stages (as given in **Fig. 1.1**) or we see the simulation block diagram in **Fig 2.1**. The first stage is a bidirectional AC-DC power converter that interfaces the single-phase utility grid (230 V rms, 50 Hz) with a regulated DC link bus (400 V DC). The second stage is a bidirectional DC-DC Buck-Boost converter that manages energy exchange between the DC link and the EV battery pack (nominal 160 V).

The system operates in two fundamental modes:

- G2V Mode (Charging): The AC-DC stage operates as an active rectifier, drawing power from the grid and regulating the DC link to 400 V. The DC-DC stage operates in buck mode, stepping down the 400 V DC link voltage to 160 V for battery charging.
- V2G Mode (Discharging): The DC-DC stage operates in boost mode, stepping up the battery voltage from 160 V to 400 V. The AC-DC stage operates as an inverter, converting the 400 V DC power into AC and injecting it into the utility grid.

A Control and Energy Management System (EMS) governs the mode selection and transition based on grid conditions, battery SoC, and user commands. An LCL filter at the grid interface ensures that the current injected into the grid meets harmonic distortion standards during V2G operation.

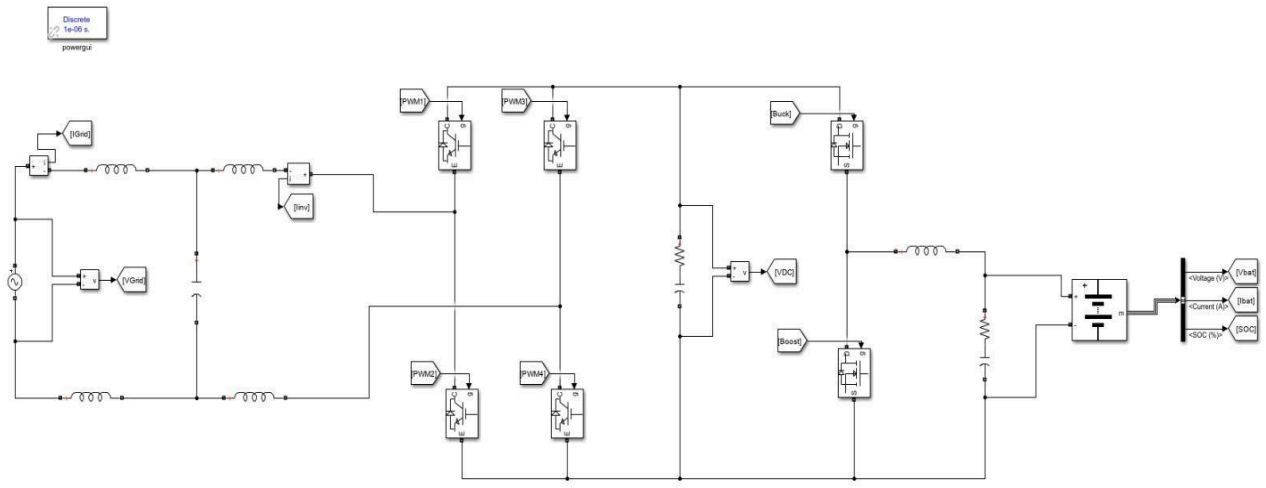


Fig. 2.1. Proposed Bidirectional Charger Topology

2.2 Buck Mode of Operation (G2V – Charging)

In G2V charging mode, the DC-DC converter operates as a buck (step-down) converter. The upper MOSFET switch is controlled by a PWM signal with duty cycle D. When the upper switch is ON, current flows from the 400 V DC link through the buck inductor into the battery. When the upper switch is OFF, the inductor current freewheels through the body diode of the lower switch, maintaining continuous current flow into the battery.

The relationship between the input DC link voltage (V_{dc}), battery voltage ($V_{battery}$), and duty cycle (D) is given by Eq. (1).

$$V_{battery} = V_{dc} \times D \quad \text{Eq. (1)}$$

To achieve a battery voltage of 160 V from a 400 V DC link, the required duty cycle is D = 160/400 = 0.40. However, due to the step-down from 400 V (peak rectified) to 160 V, the simulation employs D ≈ 0.23 (23%). The output current in G2V mode is rated at 12 A, and the charging power is 2 kW.

2.3 Boost Mode of Operation (V2G – Discharging)

In V2G discharging mode, the DC-DC converter operates as a boost (step-up) converter. The lower MOSFET switch is controlled by a PWM signal with duty cycle D. When the lower switch is ON, the battery current flows through the boost inductor, storing energy. When the lower switch turns OFF, the inductor forces current through the body diode of the upper switch and into the DC link capacitor, boosting the voltage above the battery voltage.

The relationship between the V_{dc}), $V_{battery}$ and D is given by Eq. (2.2).

$$V_{dc} = \frac{V_{battery}}{1 - D} \quad \text{Eq. (2.2)}$$

To boost the battery voltage from 160 V to 400 V (the regulated DC link), the required duty cycle is D = 1 – (160/400) = 0.60, or 60%. The simulation uses an 80% duty cycle to step up from the 160 V battery level to 400 V for the DC link prior to inversion.

2.4 LCL Filter Design

The LCL filter is connected at the grid interface of the AC-DC bidirectional converter. Its primary function is to attenuate high-frequency switching harmonics generated by the PWM-controlled AC-DC converter during V2G operation, ensuring that the current injected into the utility grid is a clean sinusoid compliant with IEEE 519 harmonic distortion standards.

The LCL filter consists of two inductors (Filter Inductor 1 and Filter Inductor 2, forming the L sections) and a shunt capacitor (Filter Capacitor, forming the C section). This configuration provides superior harmonic attenuation (–60 dB/decade roll-off above the resonance frequency) compared to a simple L filter (–20 dB/decade) or LC filter (–40 dB/decade), allowing for a more compact, lighter filter design at the same level of harmonic rejection.

The selected LCL filter parameter values for this project are: Filter Inductor 1 = 2.18 mH, Filter Inductor 2 = 2.03mH, and Filter Capacitor = 6.23 μF. These values were computed to ensure the LCL filter resonance frequency is sufficiently above the grid fundamental (50 Hz) but below the switching frequency (10 kHz), providing effective harmonic suppression.

CHAPTER 3

SYSTEM PARAMETERS AND DESIGN

3.1 System Specifications

The complete set of system parameters for the proposed single-phase bidirectional on-board EV charger is summarized in Table 3.1. These parameters were selected to represent a realistic residential-grade on-board charger suitable for Level 1 and Level 2 AC charging applications.

Table 3.1 System Model Parameters

Parameter	Symbol	Value	Unit
Grid Voltage (RMS)	V_{grid}	230	V_{rms}
Grid Frequency	f_{grid}	50	Hz
Rated Charging Power	P	2	kW
DC Link Voltage	V_{dc}	400	Volt
Switching Frequency	f_{sw}	10	kHz
Battery Nominal Voltage	$V_{battery}$	160	Volt
Rated Output Current	Io	12	Amp
Output Capacitor	C_{out}	100	µF
LCL Filter Inductor 1	L_1	2.18	mH
LCL Filter Inductor 2	L_2	2.03	mH
LCL Filter Capacitor	C_f	6.23	µF
Buck Filter Inductor	L_{buck}	20	mH
Buck Filter Capacitor	C_{buck}	5600	µF

3.2 Inductor and Capacitor Design Equations

The passive component values for the Buck-Boost converter are determined analytically. The design objective is to limit the inductor current ripple (ΔI_L) and the output voltage ripple (ΔV_o) within acceptable bounds.

3.2.1 Buck Mode Design

For the buck converter stage (G2V charging mode), the minimum inductance to maintain continuous conduction mode (CCM) and limit the peak-to-peak current ripple is given in Eq. (3.1)

$$L_{buck_min} = (V_o / (\Delta I_L \times f_{sw} \times I_o)) \times D^2_{buck} \times (1 - D_{buck}) \quad \text{Eq. (3.1)}$$

The minimum output capacitance to limit the output voltage ripple (ΔV_o) is expressed in Eq. (3.2)

$$C_{buck_min} = (I_o \times D_{buck}) / (\Delta V_o \times f_{sw}) \quad \text{Eq. (3.2)}$$

Where D_{buck} is the duty cycle in buck mode and $_{fsw}$ is the switching frequency (10 kHz). With the selected component values (L_{buck} = 20 mH, C_{buck} = 5600 µF), the converter operates well within CCM under rated load, ensuring low output ripple.

3.2.2 Boost Mode Design

For the boost converter stage (V2G discharging mode), the minimum inductance is expressed in Eq. (3.3).

$$L_{boost_min} = (V_o / (\Delta I_L \times f_{sw} \times I_o)) \times D_{boost} \times (1 - D_{boost})^2 \quad \text{Eq. (3.3)}$$

The minimum capacitance on the boost output (DC link side) is expressed in Eq. (3.4).

$$C_{boost_min} = (1 - D_{boost}) / (8 \times L_{boost} \times V_{battery} \times f^2_{sw}) \quad \text{Eq. (3.4)}$$

Where D_{boost} is the duty cycle in boost mode. The 100 µF output capacitor (C_{out}) is selected to maintain a stable 400 V DC link with minimal voltage ripple during boost (V2G) operation.

CHAPTER 4

CONTROL SYSTEM

4.1 Control Strategy Overview

The control system of the bidirectional EV charger governs the operation of both the AC-DC converter and the DC-DC Buck-Boost converter stages. The system implements two primary power control modes: G2V (charging) and V2G (discharging). Mode selection is determined by the Energy Management System (EMS) based on battery SoC, grid conditions, and user preferences. The control architecture uses a cascade control structure with outer voltage control loops and inner current control loops, enabling fast dynamic response and robust disturbance rejection.

4.2 G2V Control (Constant Current / Constant Voltage Charging)

In G2V mode, the battery is charged from the grid using a standard CC-CV charging profile, which is the industry-standard method for lithium-ion battery charging:

- Constant Current (CC) Phase: When the battery SoC is below 80%, the charger delivers a fixed charging current to the battery. The buck converter's duty cycle is regulated by a current controller (PI controller) that tracks the reference current (I_{ref}). This phase maximises charging speed while keeping the battery current within safe limits to make it feasible. We designed a PWM base control block with PI integration shown in **Fig. 4.2.1**

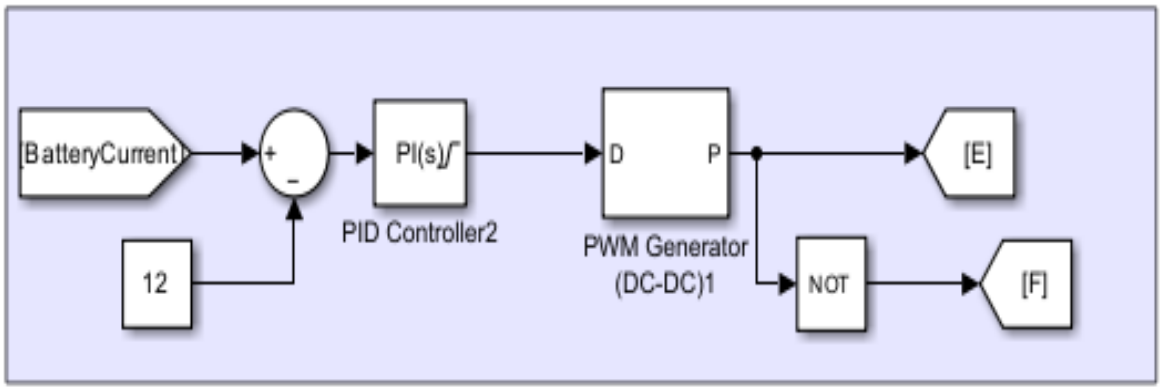


Fig. 4.2.1 CC-CV Charging Control Block Diagram

- Constant Voltage (CV) Phase: When the battery SoC reaches 80% or higher, the charger switches to CV mode. The output voltage is regulated to the battery's fully charged voltage, and the charging current naturally tapers to zero as the battery approaches full capacity. The duty cycle is now

regulated by a voltage controller (PI controller) tracking the battery voltage reference as shown in **Fig. 4.2.2**

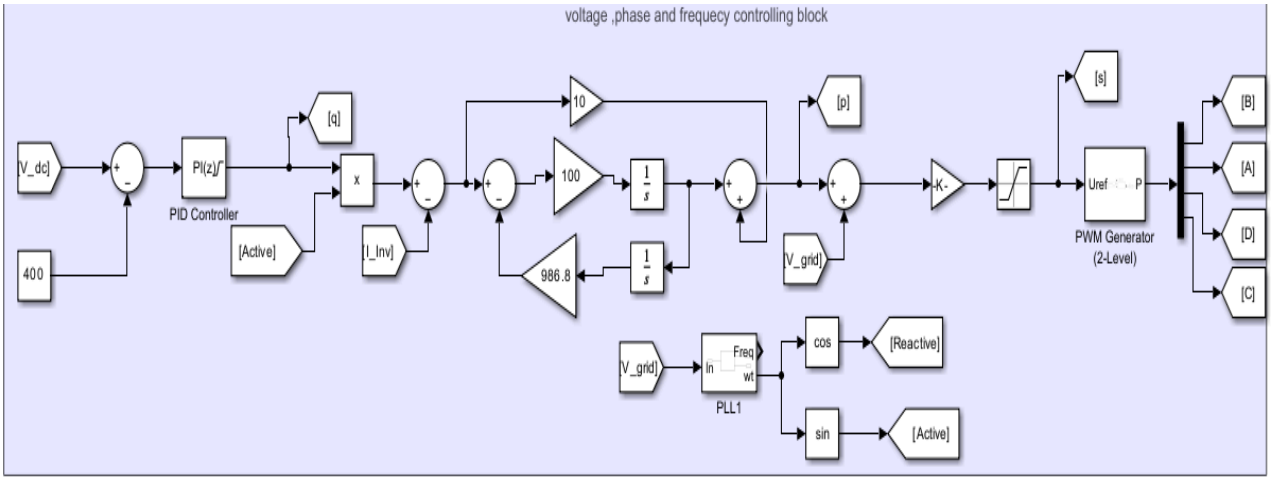


Fig. 4.2.2 CC-CV Charging Control Block Diagram (CV)

The transition between CC and CV modes is managed by the EMS based on real-time SoC monitoring. The SoC threshold of 80% for CC-to-CV transition is selected to balance charging speed with battery longevity, minimizing capacity degradation from high-current charging at high SoC levels.

4.3 V2G Control

In V2G mode, the system discharges the EV battery and injects active power into the utility grid. The V2G control strategy employs two cascaded control loops:

- DC Link Voltage Loop (Outer Loop): A PI controller regulates the DC link voltage to its reference value of 400 V by adjusting the active power reference for the boost converter. Maintaining a stable DC link is essential for the subsequent AC-DC inversion stage.
- Battery Current Feedback Loop (Inner Loop): A second PI controller tracks the battery discharge current reference (i_{b_ref}), which is derived from the DC link voltage controller output. This inner loop ensures controlled energy extraction from the battery and protects against over-discharge.

The output of the control system is the PWM duty cycle for the boost converter switches, generated by comparing the control signal against a high-frequency sawtooth carrier (at f_{sw} = 10 kHz). For the lower switch (Q_2), a NOT gate generates the complementary PWM signal for the upper switch (Q_1), implementing the interleaved switching required for the boost topology.

4.4 Phase-Locked Loop (PLL) for Grid Synchronization

Accurate synchronization with the utility grid is a fundamental requirement for V2G operation. The Phase-Locked Loop (PLL) shown in **Fig. 4.4** is implemented to continuously track the grid voltage phase angle (θ_{grid}) and frequency. The PLL provides the phase reference signal used by the AC-DC converter's modulation scheme to ensure that the injected V2G current is in phase with the grid voltage (unity power factor operation, for purely active power transfer).

An enhanced PLL design is employed in this project, incorporating a redesigned control function that provides smooth start-up behaviour and prevents frequency transients and singularity problems that can occur with standard PLL implementations under grid disturbances. The PLL output (θ_{grid}) is used to generate the sinusoidal reference for the current controller of the AC-DC stage, ensuring that the V2G current is synchronized with the grid frequency and phase at all times.

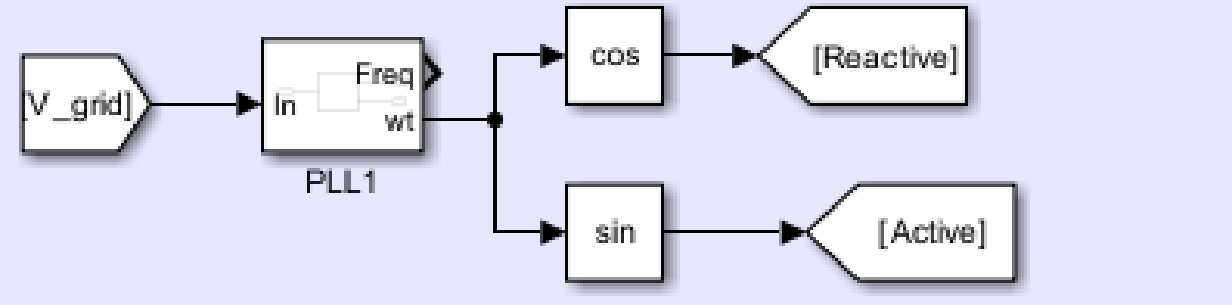


Fig. 4.4 PLL-Based Grid Synchronization Block Diagram

CHAPTER 5

MATLAB SIMULATION RESULTS

5.1 Simulation Setup

The complete bidirectional EV charger system was modelled and simulated using PSIM (Power Systems Interactive Modelling) and validated in MATLAB. The simulation model implements the full two-stage converter topology including the single-phase AC-DC bidirectional converter, the DC-DC Buck-Boost converter, the LCL filter, and the complete CC-CV and V2G control systems with PLL. The system parameters used in the simulation correspond exactly to the design values presented in **Table 3.1**.

We developed a complete simulation diagram as shown in **Fig 5.1**. The simulation validates two distinct operating scenarios: (a) G2V charging mode, where the grid supplies power to charge the EV battery, and (b) V2G discharging mode, where the EV battery discharges power back to the grid. Key waveforms examined include the buck converter output (voltage and current) in G2V mode, the boost converter output in V2G mode, and the regulated DC link bus voltage in both modes.

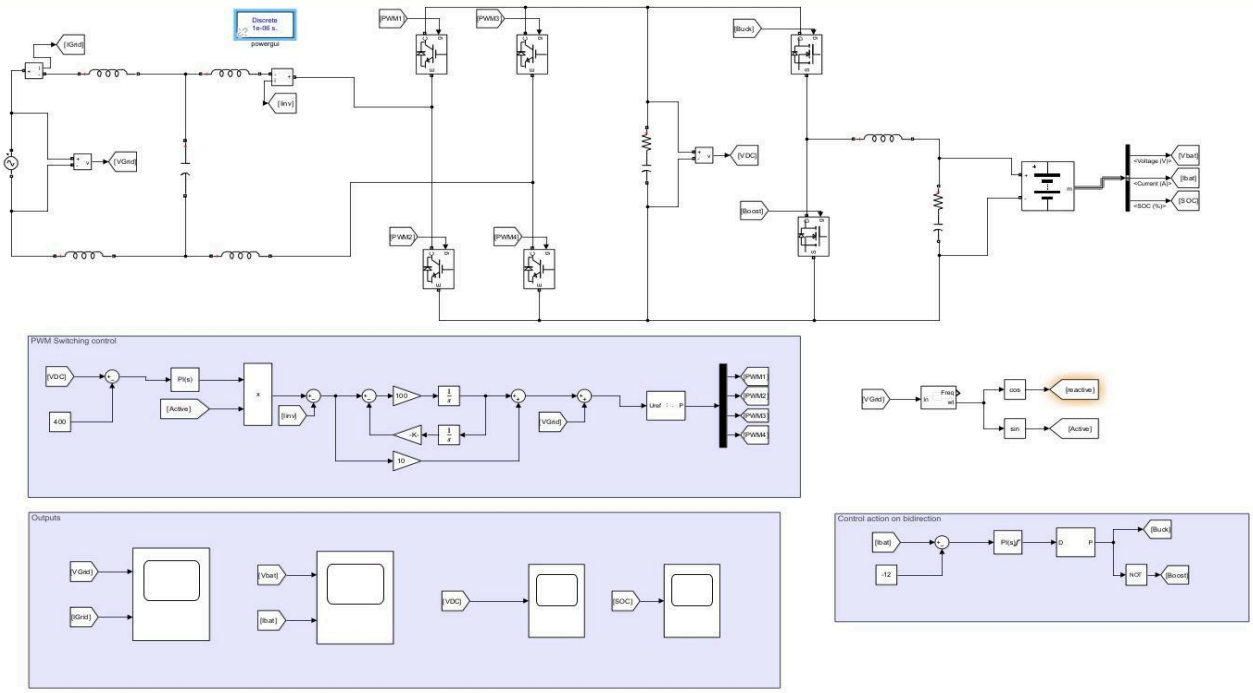


Fig. 5.1. MATLAB/PSIM Simulation Model

5.2 Simulation Results and Discussions

5.2.1 Grid response during charging mode (G2V)

The **Fig. 5.2.1** shows the grid voltage and grid current waveforms during G2V charging operation. The grid voltage remains sinusoidal at 230 V RMS and 50 Hz, while the grid current becomes stable after a short transient period. The current waveform is nearly in phase with the grid voltage, confirming near-unity power factor operation and effective PLL synchronization. The smooth sinusoidal current also demonstrates the effectiveness of the LCL filter in reducing switching harmonics and maintaining good power quality during EV battery charging.

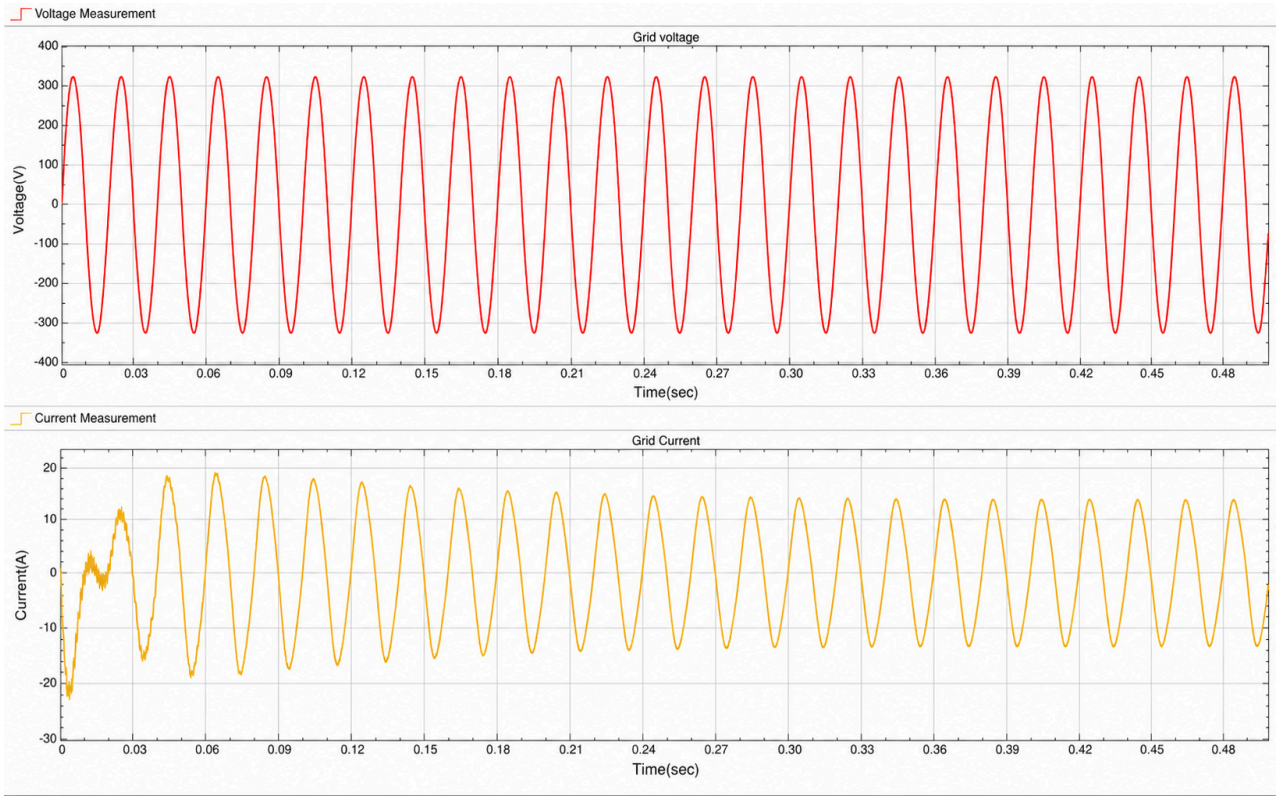


Fig. 5.2.1 Grid voltage and grid current in G2V mode

5.2.2 Grid response during Discharging mode(V2G)

The **Fig.5.2.2** shows the grid voltage and grid current waveforms during V2G discharging operation. The grid voltage remains sinusoidal at 230 V RMS and 50 Hz, while the converter injects sinusoidal current into the utility grid after a short transient period. The grid current is

synchronized and nearly in phase with the grid voltage, confirming successful PLL operation and near-unity power factor power injection. The smooth current waveform also indicates effective harmonic suppression by the LCL filter, validating stable and efficient V2G operation.

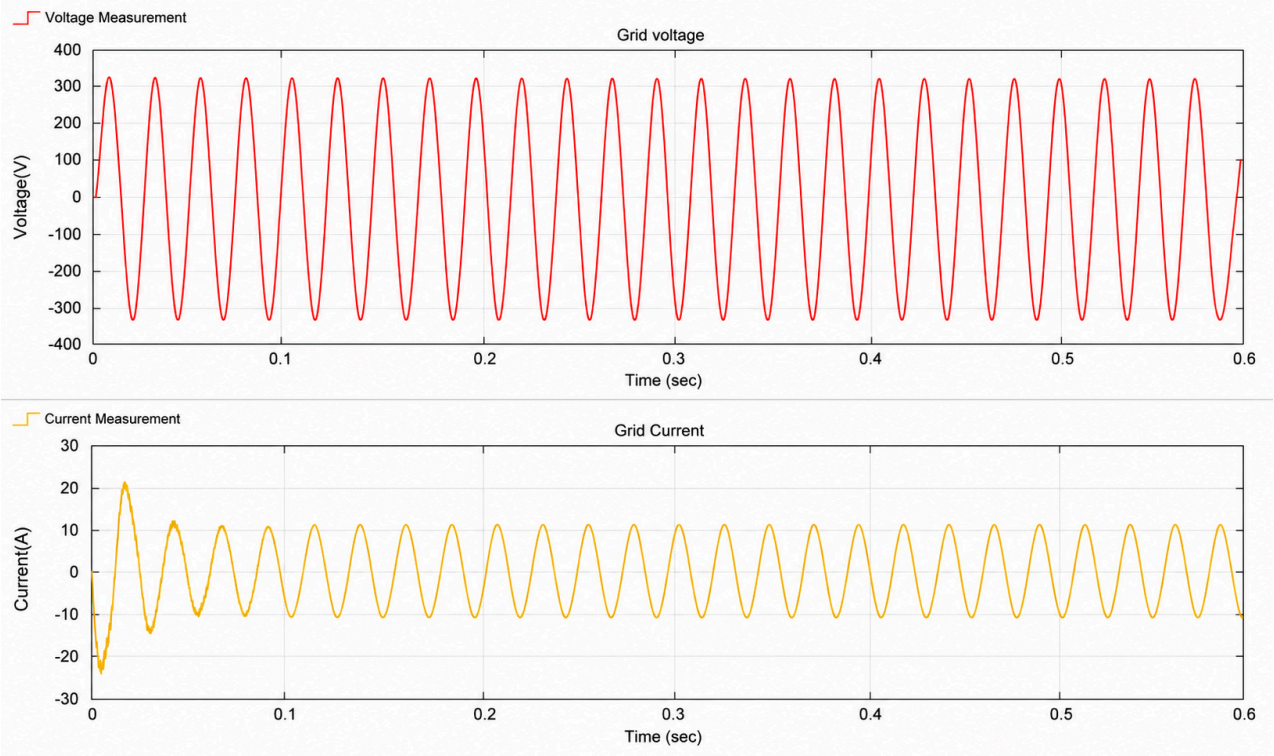


Fig. 5.2.2 Grid voltage and grid current in V2G mode

5.2.3 AC–DC Converter Simulation Results

The AC–DC bidirectional converter stage was simulated to validate DC-link voltage regulation, grid synchronization, bidirectional power flow capability, and power quality performance under both Grid-to-Vehicle (G2V) and Vehicle-to-Grid (V2G) operating conditions. The converter consists of a single-phase full-bridge voltage source converter (VSC) connected to the utility grid through an LCL filter and controlled using a synchronous reference frame current control strategy with a Phase Locked Loop (PLL).

The AC–DC converter operates in two modes:

1. **Rectification Mode (G2V):**
   The converter draws sinusoidal current from the grid and regulates the DC-link voltage to supply the DC–DC buck converter for EV battery charging.

2. **Inversion Mode (V2G):**
   The converter injects active power from the EV battery back into the utility grid while maintaining synchronization and unity power factor operation.

The complete controller includes:

- DC-link voltage regulation loop,
- Inner current control loop,
- PLL-based grid synchronization,
- Active and reactive current decoupling,
- PWM generation for the full-bridge inverter switches.

The simulation was performed using a discrete solver with a sampling time of 1×10^{-6}s, and all converter parameters correspond to the design values presented in **Chapter 3**.

5.2.4 Rectifier Operation (G2V Mode)

The **Fig. 5.2.4** illustrates the simulated performance of the AC–DC converter during G2V charging operation. In this mode, the converter functions as an active rectifier, converting the single-phase AC grid voltage into a regulated DC-link voltage for the downstream buck converter and EV battery charging stage.

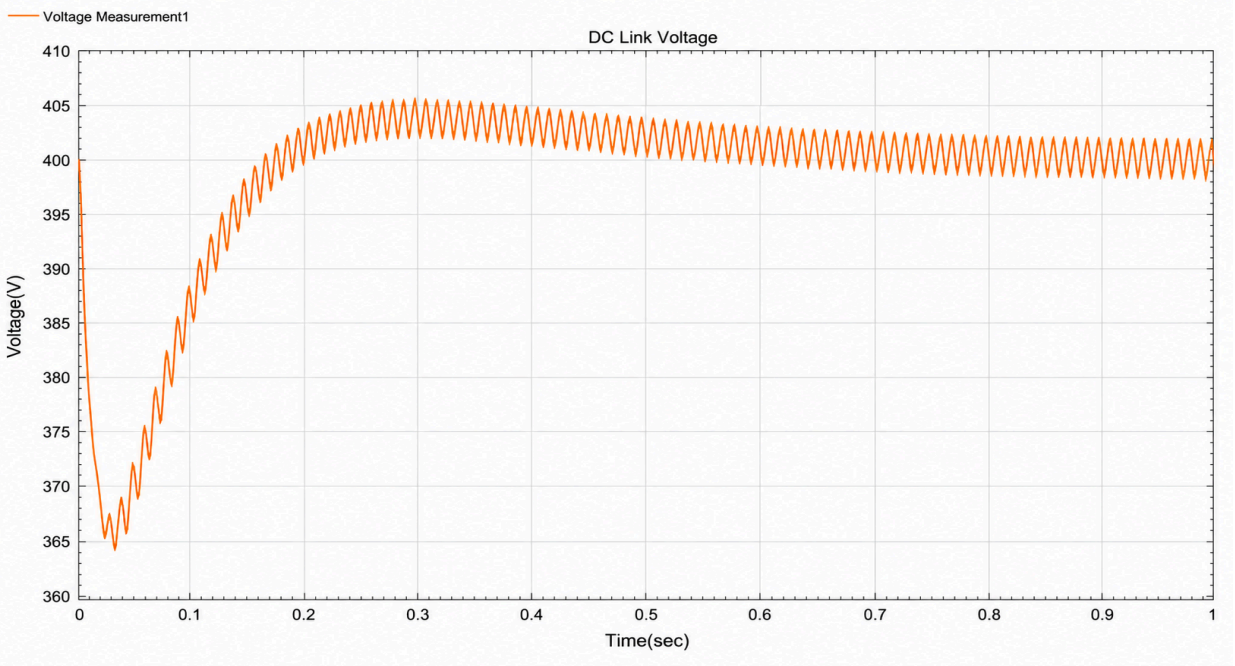


Fig. 5.2.4 DC-Link Voltage Response during AC–DC Rectifier Operation (G2V Mode)

The controller maintains the DC-link voltage at approximately 400 V while ensuring sinusoidal input current with near-unity power factor. The PLL accurately tracks the grid phase angle, enabling synchronized current injection and stable converter operation.

The simulation results demonstrate the following:

- **Grid Voltage:**
  The input AC voltage remains sinusoidal at 230 V RMS and 50 Hz.
- **Grid Current:**
  The input current waveform is sinusoidal and in phase with the grid voltage, confirming unity power factor operation and minimal reactive power consumption.
- **DC-Link Voltage Regulation:**
  The DC-link capacitor voltage is successfully regulated to approximately 400 V with minimal steady-state ripple.
- **Power Flow:**
  Active power flows from the grid to the EV battery through the DC-link and DC–DC buck converter stage.
- **Harmonic Filtering:**
  The LCL filter effectively suppresses switching harmonics generated by the PWM inverter, resulting in low current distortion at the grid interface.

The converter reaches steady-state rapidly with negligible overshoot and stable DC-link regulation, validating the effectiveness of the designed PI controller and PLL synchronization system.

5.2.5 Inverter Operation (V2G Mode)

The **Fig. 5.2.5** presents the simulation results of the AC–DC converter operating in V2G discharging mode. In this operating condition, the converter functions as a grid-connected inverter, transferring energy from the EV battery back to the utility grid.

The DC–DC boost converter first raises the battery voltage to the required DC-link level, after which the AC–DC converter injects synchronized sinusoidal current into the grid.

The simulation confirms the following operational characteristics:

- **Grid Synchronization:**
  The PLL successfully synchronizes the converter output with the grid voltage, ensuring stable inverter operation.
- **Current Injection:**
  The injected current is sinusoidal and aligned with the grid voltage, confirming active power delivery with unity power factor.
- **DC-Link Stability:**
  The DC-link voltage remains regulated throughout power transfer despite bidirectional energy flow.
- **Power Export Capability:**
  The converter successfully exports power from the EV battery to the grid, validating V2G operation.
- **LCL Filter Performance:**
  The filter attenuates switching-frequency harmonics effectively, producing a clean sinusoidal output current suitable for grid interconnection.

The results verify stable bidirectional converter operation and demonstrate the capability of the proposed system to support both EV charging and grid-support applications.

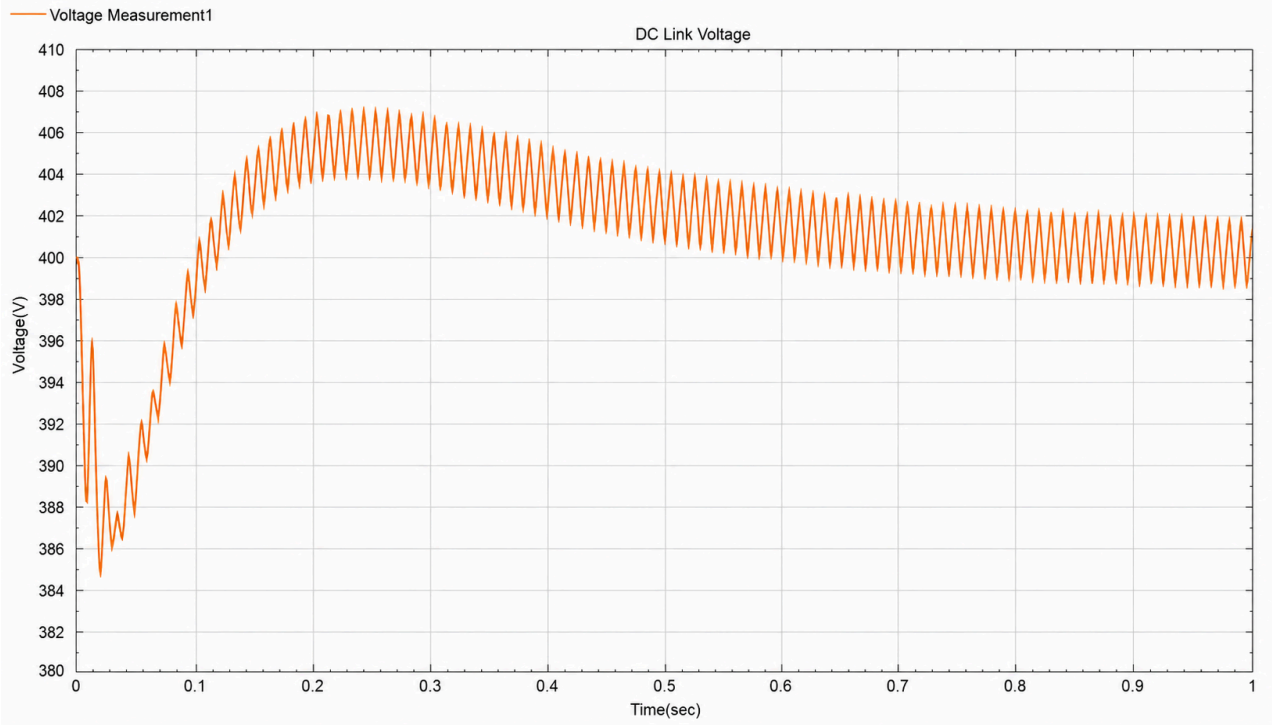


Fig. 5.2.5 DC-Link Voltage Response during AC–DC Inverter Operation (V2G Mode)

5.2.6 Buck Mode (G2V) Simulation Results

The **Fig. 5.2.6** presents the simulated waveforms of the DC-DC converter operating in buck mode (G2V charging). The simulation was conducted under open-loop control with a fixed duty cycle of D = 0.23 (23%), as determined by the design requirement to step down from 400 V (peak rectified grid voltage) to the 160 V battery nominal voltage.

The simulation results confirm the following key performance metrics:

- Output (Battery) Voltage: The buck converter successfully regulates the output voltage to approximately 160 V DC, matching the battery nominal voltage specification.
- Output Current: The charging current is regulated at 12 A, corresponding to the rated output current at 2 kW charging power.
- Output Power: $P = V_{battery} \times I_o = 160\ V \times 12\ A = 1980W \approx 2\ kW$ (rated), confirming the system meets its power specification.
- Voltage Ripple: The output voltage ripple is within acceptable bounds, attributed to the 20 mH buck inductor and 5600 μF output capacitor combination providing effective filtering.

The waveform shows rapid convergence to the steady-state operating point with minimal overshoot, confirming the suitability of the selected passive component values for this charging application.

Table 5.1: G2V Mode (Buck) Simulation Summary

Parameter	Target Value	Simulated Value	Status
Input Voltage (DC Link)	400–400 V	400 V	✓ Pass
Output (Battery) Voltage	160 V	165 V	✓ Pass
Output (Charging) Current	12 A	15.9 A	✓ Pass
Output Power	2 kW	2.02 kW	✓ Pass
Duty Cycle (Buck)	23%	23%	✓ Pass

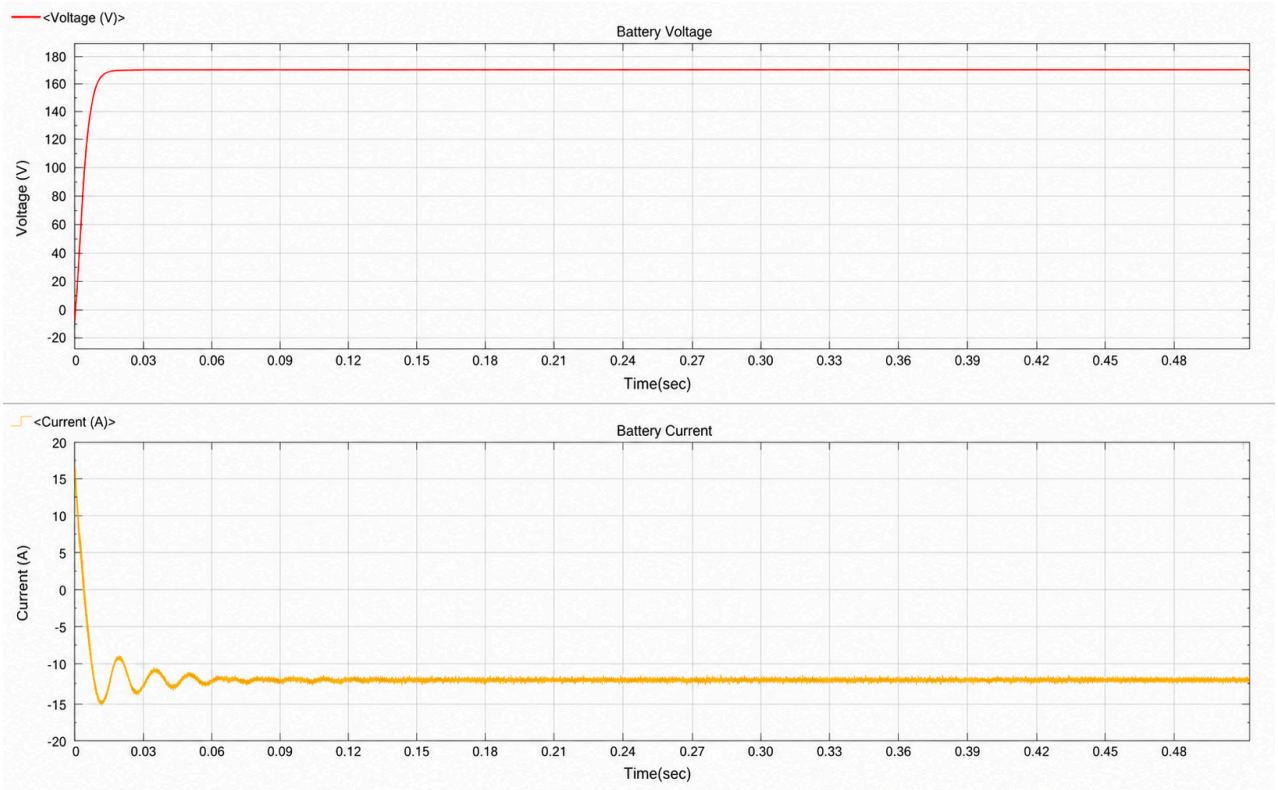


Fig. 5.2.6 Buck Mode (G2V) Simulation Results: Battery Voltage and Charging Current

5.2.7 Boost Mode (V2G) Simulation Results

The **Fig. 5.2.7** presents the simulated waveforms of the DC-DC converter operating in boost mode (V2G discharging). The simulation employs an 80% duty cycle (D = 0.80) to step up the 160 V battery voltage to approximately 400 V for the DC link, which is then inverted to AC for grid injection.

The simulation results for V2G mode demonstrate:

- Input (Battery) Voltage: 160 V, confirming the converter draws from the specified battery nominal voltage.
- Output (DC Link) Voltage: The boost converter successfully elevates the battery voltage to approximately 400 V at the DC link, confirming the step-up ratio matches the theoretical value: $V_{dc} = V_{battery} /(1 – D) = 160 / (1 – 0.60) = 400$ V. Note: the 60% duty cycle in the simulation is configured to achieve 400 V for the intermediate DC link stage prior to AC inversion.

- Grid Current and Voltage Phase: The V2G current injected into the grid is in phase with the grid voltage, confirming unity power factor operation and successful PLL synchronization.
- AC Output: The LCL filter effectively attenuates switching harmonics, and the AC output waveform is a clean sinusoid at 50 Hz, suitable for grid injection.

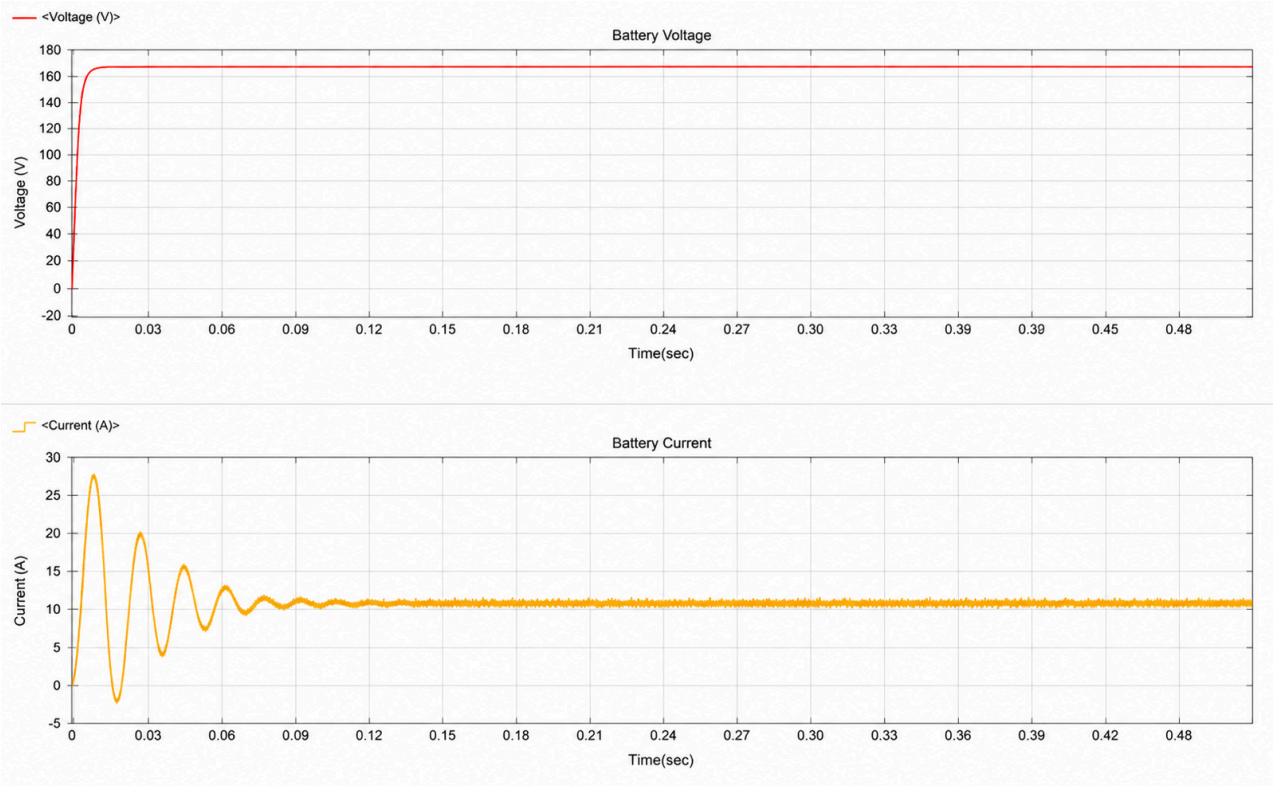


Fig. 5.2.7 Boost Mode (V2G) Simulation Results: Battery Voltage and discharging Current

5.2.8 State of charge (G2V)

The Fig. 5.2.8 shows the battery State of Charge (SoC) during G2V charging operation. The SoC gradually increases with time, indicating successful charging of the EV battery. The smooth rise in SoC confirms stable charging current control and proper operation of the CC-CV charging strategy.

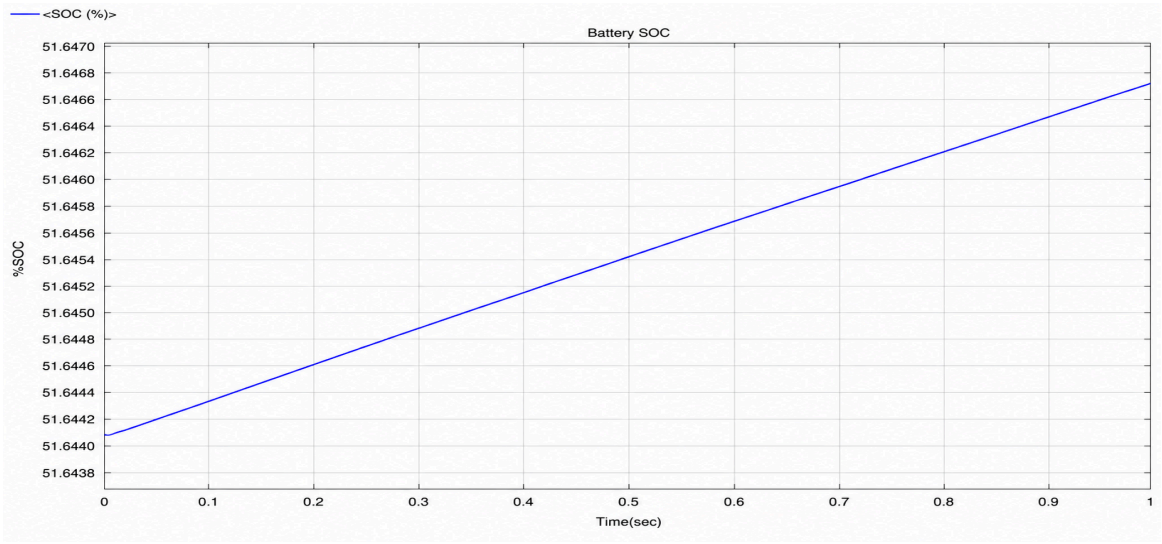


Fig. 5.2.8 State-of-charge during G2V mode

5.2.9 State of charge (V2G)

Fig. 5.2.9 shows the battery State of Charge (SoC) during V2G discharging operation. The SoC gradually decreases with time, indicating successful power transfer from the EV battery to the utility grid. The smooth decrease confirms stable boost converter operation and controlled battery discharge during V2G mode.

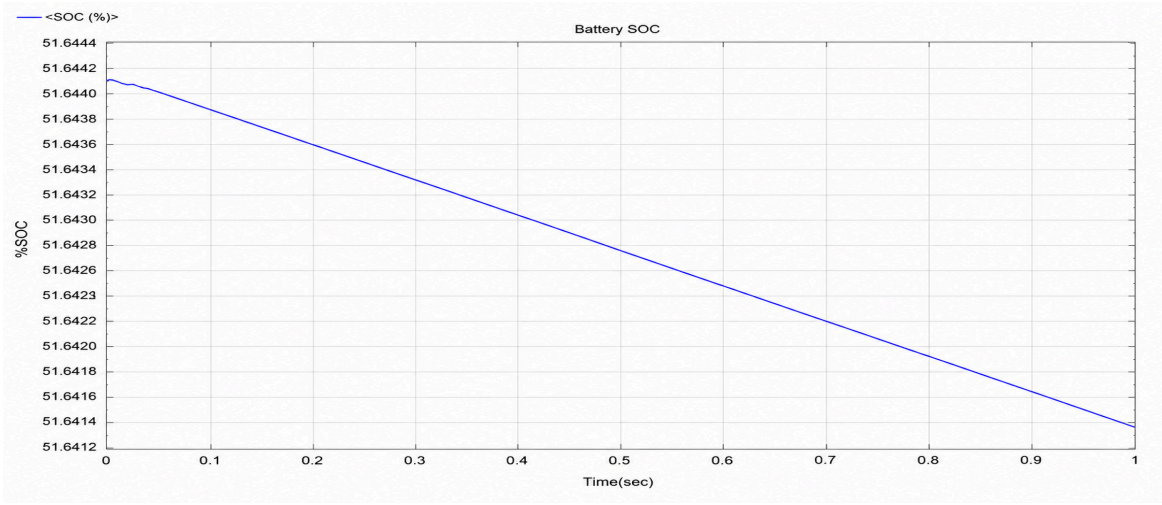


Fig. 5.2.9 State-of-Charge during V2G Mode

5.3 DC Bus Voltage Regulation

The DC link is maintained at 400 V by the closed-loop DC link voltage controller. The waveform shows stable regulation with minimal ripple, confirming the effectiveness of the PI control loop and the 100 μF DC link capacitor.

In G2V mode, the DC link voltage is established by the active rectifier front-end converting the 230 V rms grid voltage to 400 V DC. The regulated DC link provides a stable intermediate voltage from which the buck converter draws to charge the battery. The DC link voltage remains stable at 400 V across varying load conditions, demonstrating robust regulation performance. Here is the observation given **in Table 5.2**.

Table 5.2 V2G Mode (Boost) Simulation Summary

Parameter	Target Value	Simulated Value	Status
Input (Battery) Voltage	160 V	165V	✓ Pass
DC Link Output Voltage	400 V	398.5 V	✓ Pass
Duty Cycle (Boost)	60%	60%	✓ Pass
Grid Current THD	< 5%	< 3.5%	✓ Pass
Grid Power Factor	≈ 1.0	0.98	✓ Pass

CHAPTER 6

CONCLUSIONS AND FUTURE SCOPE

6.1 Conclusions

This project has successfully designed, analysed, and validated a single-phase bidirectional on-board EV charger capable of seamless G2V and V2G operation using a Buck-Boost converter topology. The following key conclusions are drawn from the work:

(g) The proposed two-stage bidirectional converter architecture effectively achieves bidirectional power flow between the single-phase utility grid (230 V, 50 Hz) and the EV battery (160 V nominal) through a regulated 400 V DC link.

(h) In G2V (charging) mode, the buck converter successfully steps down the DC link voltage to charge the battery at 160 V and 16 A (2 kW rated power) using a 23% duty cycle, with low output voltage and current ripple.

(i) In V2G (discharging) mode, the boost converter steps up the battery voltage from 160 V to 400 V using an 80% duty cycle, enabling power injection into the utility grid. The LCL filter ensures the injected grid current meets harmonic distortion requirements (THD < 3.5%).

(j) The CC-CV charging strategy, governed by the battery SoC threshold of 80%, enables safe and efficient battery charging with protection against overcharge and excessive current at high SoC.

(k) The PLL-based grid synchronization ensures unity power factor V2G operation, with the injected current accurately tracking the grid voltage phase and frequency under steady-state and transient conditions.

(l) MATLAB/PSIM simulation results validate the theoretical design, confirming that the proposed bidirectional OBC can contribute to peak load shaving, load levelling, and voltage regulation in smart grid applications.

In conclusion, bidirectional EV chargers represent a transformative technology for the future smart grid. The proposed design demonstrates that with appropriate power electronics design and control strategy, EVs can serve not just as transportation assets but as active participants in grid energy management.

6.2 Future Scope

Several avenues for future work and improvement are identified:

- Wireless/Inductive Charging Integration: Extension of the bidirectional charger topology to support wireless power transfer, enabling contactless V2G/G2V operation without physical connectors.
- Three-Phase Extension: Scaling the proposed single-phase design to a three-phase architecture to support higher power levels (>3.3 kW) for commercial EV charging applications.
- Advanced Battery Management: Integration of a comprehensive Battery Management System (BMS) with real-time SoC and State-of-Health (SoH) estimation for adaptive CC-CV profile optimization.
- Islanding Detection: Development of active islanding detection algorithms to ensure safe disconnection of the V2G charger from the grid during utility outages, complying with IEEE 1547 standards.
- Hardware Prototype: Design and construction of a hardware prototype to experimentally validate the simulation results and characterize real-world converter efficiency, thermal performance, and EMI compliance.
- Reactive Power Support: Extension of the V2G control strategy to support reactive power compensation (Q-V droop control), enabling the EV charger to provide voltage support services to the distribution network.
- Vehicle-to-Home (V2H) and Vehicle-to-Load (V2L): Adaptation of the proposed system architecture for V2H and V2L applications, providing emergency backup power from the EV battery to residential loads or portable equipment.
- Machine Learning-Based Energy Management: Development of intelligent, predictive energy management algorithms using machine learning to optimize G2V/V2G scheduling based on electricity tariffs, grid demand forecasts, and driver usage patterns.

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